

Design CHAT Application

① Requirement Gathering

Functional

- I send / receive msg
- Group msg support
- Last seen
- User login / Authentication

Non-functional

- Scalability → handle huge traffic
- Low latency → send or receive should fast
- Availability

② Back of the Envelope:-

Total users ⇒ 2 Billion
 DAU :- 50 million
 1 user → 10 msg → 4 people in 1 day
 Now many msg per day ⇒ 50 million × 40 msg
 2000 million
 ⇒ 2 Billion msg/day

1 msg = 100 byte
 (100 character)

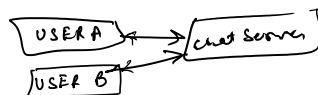
2 billion ⇒ 2 Billion × 100 byte
 ⇒ 200 GB / day

for 10 years
 ⇒ 200 GB × 365 × 10
 ⇒ 730 000 GB
 ⇒ 730 TB

③ Design



If IP address known then they can talk to each other P2P model but it's not scalable
 So now we need that server so it will not be P2P, it's Client Server model

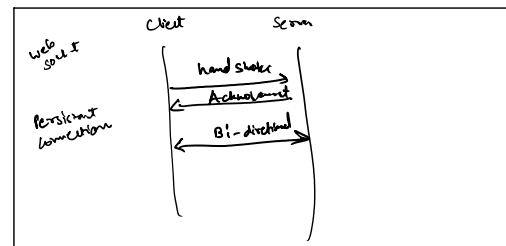
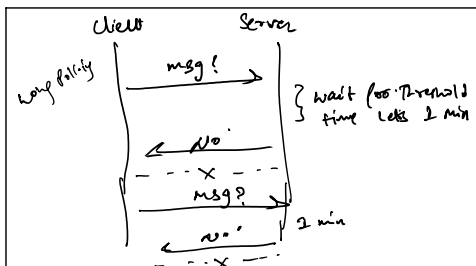
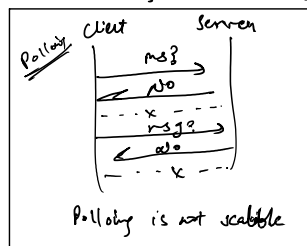


Now what Protocol should we use?

If HTTP :- its Request Response, so how server will tell to User B so it will not work

there are other ways also Polling, Long Polling (Pushing) and websocket

HTTP
 Polling → Long Polling



for our server we have to use websocket



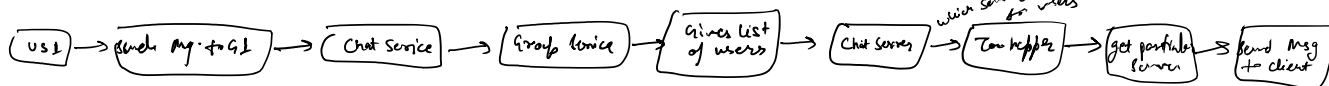
How does server know User 1 wants to talk to User 2

⇒

for this there is one more service called

user mapping service / user detector service or you can say zookeeper





Last Seen

